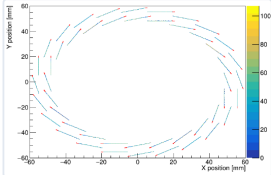
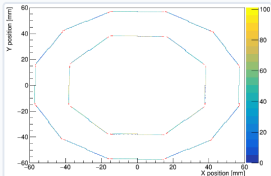
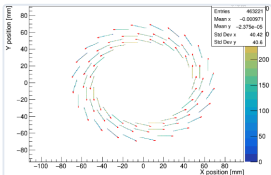
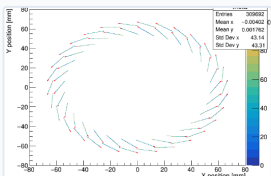
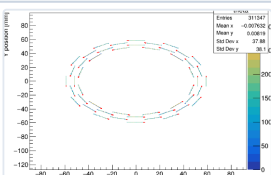
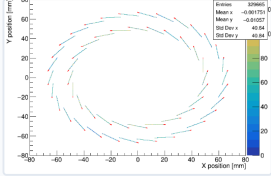
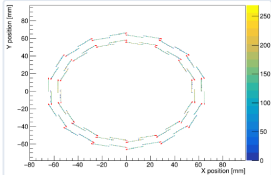
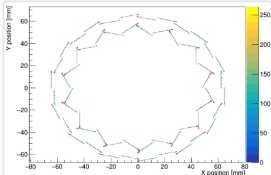
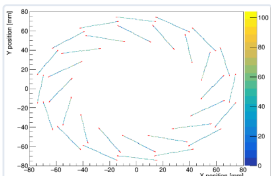


R3BRoot Simulated Barrel Geometries

Comprehensive comparative analysis of detector layout parameters, efficiency, and vertex resolution

HICS	GEOMETRY LAYOUT	INNER BARREL RADIUS (CM)	OUTER BARRELS RADIUS (CM)	INNER BARREL TILT ANGLE	OUTER BARRELS TILT ANGLE	EFFICIENCY*	VERTEX RESOLUTION (Σ_z IN MM)	COMMENTS
38		3.8	5.7	0°	0°	54%	0.14	In this configuration each module has one bigger cooling plate where two HICs are glued together horizontally.
48		5.0	5.7	15°	15°	1 st : 69% 2 nd : 58%	1 st : 0.173 2 nd : 0.172	In this configuration each module has two HICs sharing vertically the same cooling pipe.
72		5.0	2 nd barrel: 5.7 3 rd barrel: 6.7	15°	2 nd barrel: 15° 3 rd barrel: 15°	1 st : 84% 2 nd : 71%	1 st : 0.1755 2 nd : 0.1752	More than the planned 60 HICs.
60		5.79	6.47	32.15°	16°	1 st : 75.5% 2 nd : 69%	1 st : 0.1998 2 nd : 0.20	The vertex resolution varies between 0.4 and 0.2 mm depending on the hit local position in the ALPIDE.
48		1 st barrel: 4.845 2 nd barrel: 5.195	3 rd barrel: 5.545 4 th barrel: 5.895	0°	0°	1 st : 71% 2 nd : 62%	1 st : 0.172 2 nd : 0.170	The vertex resolution remains constant independently of the hit local position in the ALPIDE.

HICS	GEOMETRY LAYOUT	INNER BARREL RADIUS (CM)	OUTER BARRELS RADIUS (CM)	INNER BARREL TILT ANGLE	OUTER BARRELS TILT ANGLE	EFFICIENCY*	VERTEX RESOLUTION (Σ_Z IN MM)	COMMENTS
54		5.0	6.47	15°	15°	1 st : 79.5% 2 nd : 67.2%	1 st : 0.169 2 nd : 0.167	The vertex resolution varies between 0.25 and 0.15 mm depending on the hit local position in the ALPIDE. For two HICs to use the same cooling pipe, they will have to share it horizontally instead of vertically. This makes the cooling design harder.
56		1 st barrel: 5.474 2 nd barrel: 5.673	3 rd barrel: 6.290 4 th barrel: 6.488	0°	0°	1 st : 71.5% 2 nd : 71.2%	1 st & 2 nd : 0.187	The vertex resolution varies between 0.3 and 0.18 mm depending on the hit local position in the majority of the ALPIDE. However, at the right end of the ALPIDE it gets much worse to 0.5 mm. The overlapping of active layer together with the ghost areas make this geometry not ideal. 4 different modules share the same cooling pipe in this configuration.
56		1 st barrel: 5.201 2 nd barrel: 5.400	3 rd barrel: 6.290 4 th barrel: 6.488	1 st barrel: 25.14° 2 nd barrel: -25.14°	0°	1 st : 74.2% 2 nd : 70.98%	1 st : 0.185 2 nd : 0.184	4 different modules share the same cooling pipe in this configuration.
64		5.7026	7.1952	34°	8°	1 st : 69.7% 2 nd : 64.5%	1 st : 0.195 2 nd : 0.194	Each module has two HICs placed laterally, like in the first design.

***Efficiency Calculation Conditions:**

- **1st Condition:** Requires only that all four hits occurred within the active layer of the ALPIDE sensor.
- **2nd Condition:** Adds the constraint that the hits must not have passed through any dead layer of the inner barrel.